UTS Communication Protocol

Case 2: Product/Inventory API Analysis

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Tujuan & Skenario Bisnis

🎯Tujuan

Verifikasi komunikasi client-server melalui HTTP protocol, observasi headers/status code, dan analisis JSON data encoding pada sistem inventaris.

💼Skenario

Simulasi manajemen stok barang: Melakukan penarikan data (GET) serta penambahan produk baru (POST) pada database backend melalui API.

Endpoint & Design Request

- **GET All:** /api/products - Menampilkan seluruh daftar inventaris.
- **GET Detail:** /api/products/1 - Menampilkan spesifik produk ID 1.
- **POST Data:** /api/products - Injeksi payload JSON (Smart Sensor & NodeMCU).
- **Headers:** Menggunakan Content-Type: application/json untuk validasi POST.

Evidence GET & POST

Eksekusi Sukses

Seluruh request dieksekusi via PowerShell menggunakan curl.exe.

- GET: Menghasilkan HTTP 200 OK.
- POST: Menghasilkan HTTP 201 Created.
- Data dikembalikan dalam format terstruktur (JSON).

```
PS C:\Users\warda> curl.exe -i -X GET "http://localhost:8088/api/products"
HTTP/1.0 200 OK
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:36:07 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 642
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
```


```
{
  "resource": "products",
  "case": "Product",
  "count": 3,
  "methods": [
    "GET",
    "POST",
    "PUT",
    "PATCH",
    "DELETE"
  ],
  "canonicalPath": "/api/products",
  "itemPathExample": "/api/products/1",
  "data": [
    {
      "id": 1,
      "name": "Laptop Pro 14",
      "category": "computer",
      "price": 14500000,
      "stock": 5
    },
    {
      "id": 2,
      "name": "Temperature Sensor DHT22",
      "category": "iot",
      "price": 85000,
      "stock": 30
    },
    {
      "id": 3,
      "name": "Wireless Router AC1200",
      "category": "networking",
      "price": 475000,
      "stock": 10
    }
  ]
}
```


```
PS C:\Users\warda>
```

Evidence GET & POST

```
PS C:\Users\warda> curl.exe -i -X POST "http://localhost:8088/api/products
" -H "Content-Type: application/json" -d '{\"name\": \"Smart Sensor\", \"c
ategory\": \"iot\", \"price\": 175000, \"stock\": 12}'
HTTP/1.0 201 Created
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:48:04 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 137
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Location: /api/products/4

{
  "id": 4,
  "name": "Smart Sensor",
  "category": "iot",
  "price": 175000,
  "stock": 12,
  "createdAt": "2026-06-07T19:48:04+0700"
}
PS C:\Users\warda>
```

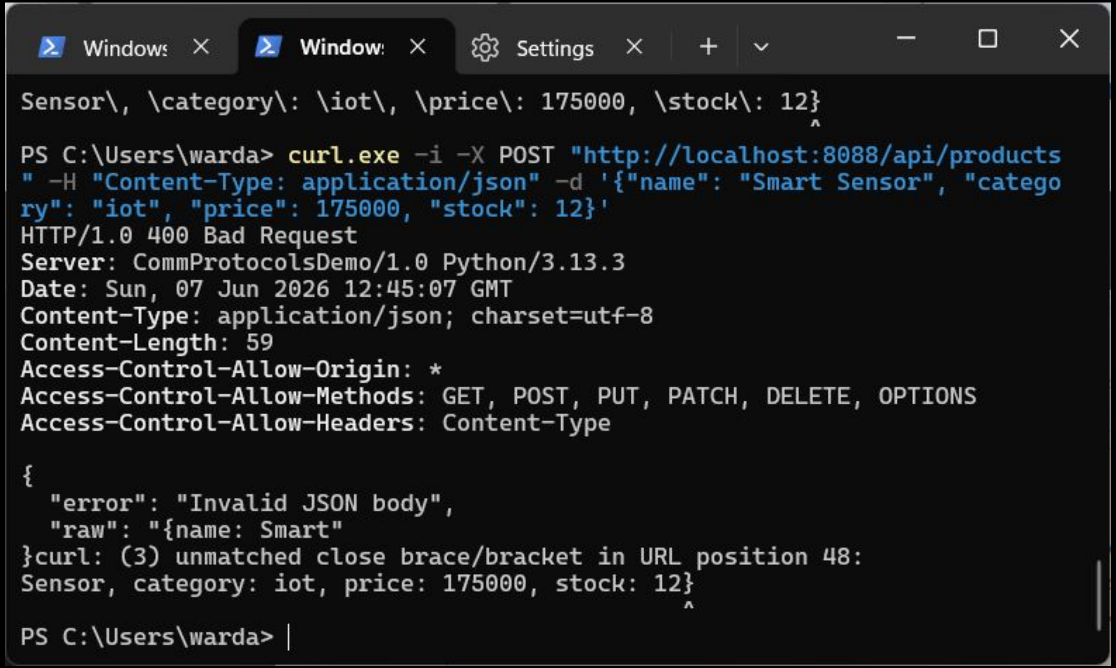
```
}
PS C:\Users\warda> curl.exe -i -X GET "http://localhost:8088/api/products/
1"
HTTP/1.0 200 OK
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:39:13 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 101
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type

{
  "id": 1,
  "name": "Laptop Pro 14",
  "category": "computer",
  "price": 14500000,
  "stock": 5
}
PS C:\Users\warda>
```

```
PS C:\Users\warda> curl.exe -i -X POST "http://localhost:8088/api/products
" -H "Content-Type: application/json" -d '{\"name\": \"Microcontroller Nod
eMCU\", \"category\": \"iot\", \"price\": 65000, \"stock\": 25}'
HTTP/1.0 201 Created
Server: CommProtocolsDemo/1.0 Python/3.13.3
Date: Sun, 07 Jun 2026 12:49:02 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 147
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type
Location: /api/products/5

{
  "id": 5,
  "name": "Microcontroller NodeMCU",
  "category": "iot",
  "price": 65000,
  "stock": 25,
  "createdAt": "2026-06-07T19:49:02+0700"
}
PS C:\Users\warda>
```

Kendala & Solusi



Solusi: Escape String

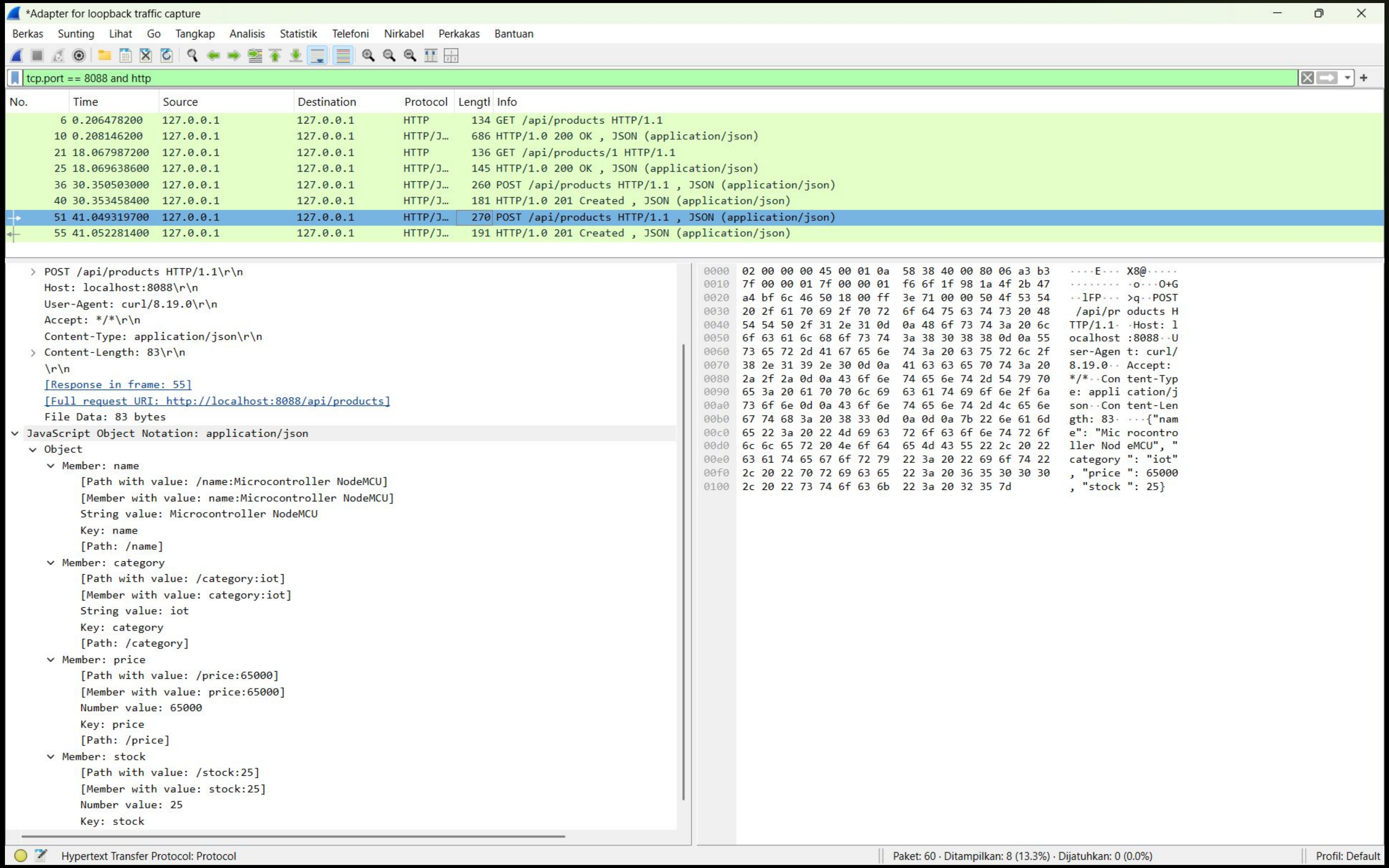
Menggunakan karakter pelolosan \" pada setiap key dan value JSON untuk menjaga integritas payload saat transmisi.

```
-d '{\"name\": \"Smart Sensor\"...}'
```

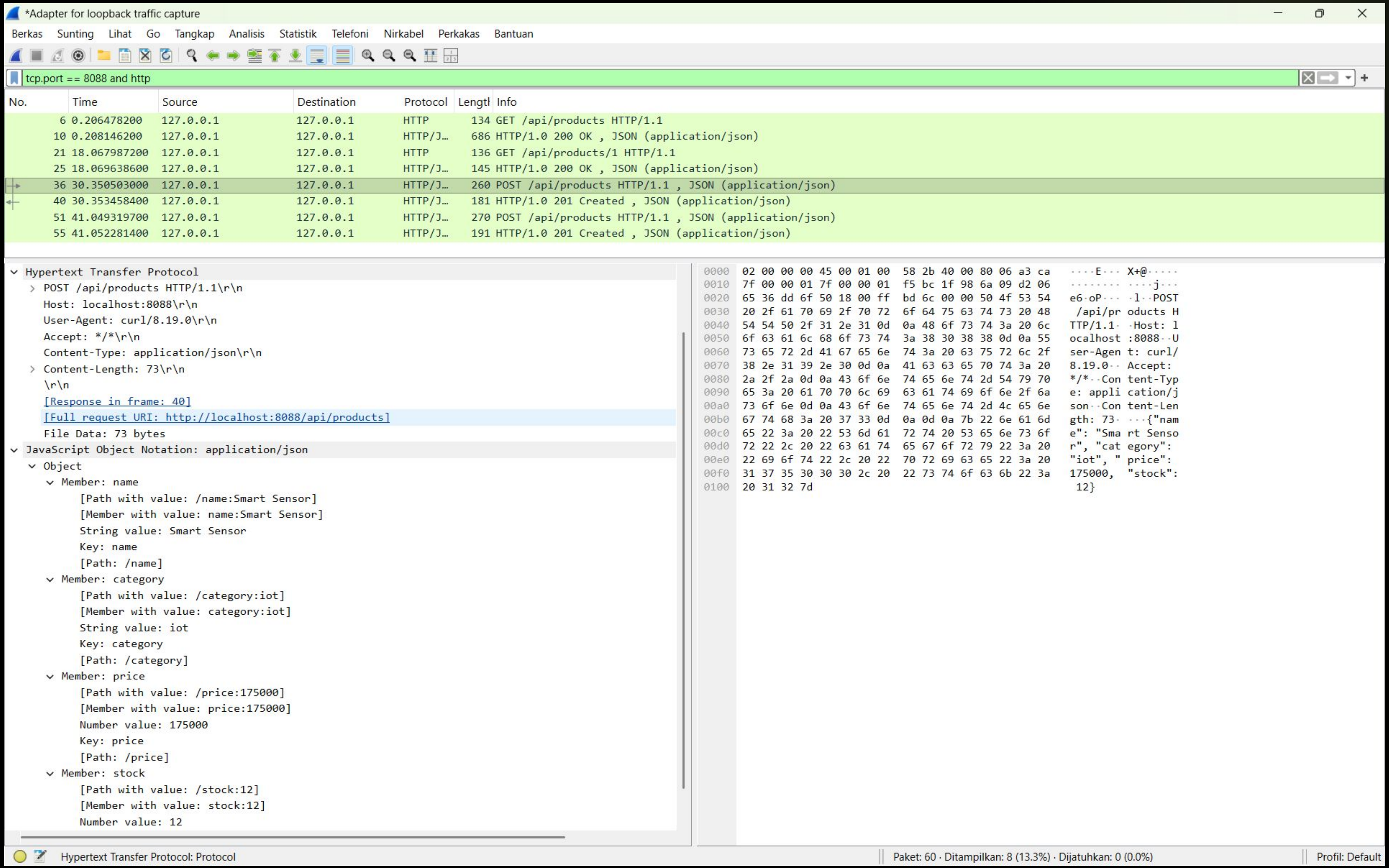
Analisis Response

Komponen	Hasil Analisis Teknis
Status Code	200 OK (GET) & 201 Created (POST Berhasil).
Encoding	JSON terenkode UTF-8 (Header: application/json).
Data Types	String (name, cat) & Integer (id, price, stock).
CORS	Access-Control-Allow-Origin: * (Izin publik).

Evidence Traffic Capture Wireshark



Evidence Traffic Capture Wireshark



Analisis TRAFFIC CAPTURE (WIRESHARK)

Komponen	**Hasil Analisis Teknis**
Loopback Capture	Pemantauan *traffic* komunikasi lokal (127.0.0.1) pada *port* 8088.
Request-Response Sinkron	Paket HTTP berjalan lancar di atas protokol TCP tanpa deteksi *packet loss* atau anomali jaringan.
Inspeksi JSON Dissector	Wireshark berhasil membedah objek *request* POST. *Payload* data "Smart Sensor" dan "Microcontroller NodeMCU" terbukti terserialisasi secara utuh.
Validasi Resolusi	Membuktikan secara konkret bahwa implementasi *escape string* (\") berhasil menjaga integritas format JSON saat transmisi.

TERIMA KASIH

Kesimpulan: Komunikasi client-server via REST API sukses diimplementasikan dan dianalisis.

https://github.com/AdityaaWardanaa/communication-protocol-uts